Neeraj Nagar

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Education

University of Waterloo, Canada — Sep 2023 - Aug 2025
MASc (thesis, computer software) in Electrical and Computer Engineering — Waterloo, Ontario
Scholarship: Graduate research studentship, International Master's Award of Excellence (IMAE)

Indian Institute of Technology (IIT) BHU, India — Jul 2016 - May 2020
Bachelor of Technology in Electronics Engineering — Varanasi, India

Experience

Software Engineer, Camera Team, Samsung R&D – India — Feb 2021 – Jul 2023

- Developed and optimized camera software for Samsung smartphones, focusing on image stabilization, deblurring, and macro photography. This improved user experience and elevated image quality across multiple models.
- Optimized the hyper-lapse video stabilization feature on flagship Samsung devices, delivering smoother, more stable video capture, strengthening the product's competitive edge in the market.
- Enhanced low-light photography performance by 20% by optimizing camera processing algorithms, reducing noise and improving image clarity in challenging lighting conditions, thereby boosting overall camera efficiency.
- Integrated camera firmware updates for Android 12 & 13 on Galaxy S21 and S22, ensuring compatibility and performance.

Software Engineer Intern, Samsung R&D – India — May 2019 – Jul 2019

- Developed a model using OpenCV and TensorFlow to transfer style between images based on semantic categories.
- Optimized style transfer algorithms, boosting efficiency, accuracy, and ensuring high-quality output with faster processing.
- Trained a model on MIT ADE20K, optimizing 27 classes for faster segmentation and improved performance.

Publications

Methods and Systems for Enhancing an Image (Patent) — 2021
Published, First Examination Report (FER) Issued Application no. 202211002117

Enhanced Adaptive Equalization based on User Behavior using Clustering (Research Paper) — 2022
Published in IJACEN, Volume-10, Issue-10, DOI

System and Method for Generating a Wide Field of View Image (Patent) — 2022
Published, Under Examination, Application no. 202211072053

System and Method for Frame Selection and Sensors Data Correction for Image Deblurring (Patent) — 2023
Published, Under Examination, Application no. 202311044055

Enhancement of Macrophotography for Mobile Devices (Patent) — 2023
Request for Examination (RQ) Filed, Application no. 202311062053

Projects

Pattern Mining of Time Series Data

- Developed a novel method to identify distinct and recurring patterns in a given time series. The primary focus of this approach is to ensure high coverage of the time series.
- Used Stumpy library to compute the matrix profile and MiniZinc framework for constraint solving.

Emergency Vehicle Detection Using YOLOv8 for Enhanced Driver Assistance

- Developed a real-time emergency vehicle detection system using YOLOv8, improving siren identification accuracy to 78%.
- Labeled and augmented 2600+ images, enhancing model robustness and real-world detection performance.
- Optimized YOLOv8 for real-time use in resource-limited environments, ensuring high-speed and efficient detection.

Autonomous Vehicle with Image Processing

- Developed an autonomous vehicle using color segmentation for object detection and Dijkstra's algorithm for optimized path planning, integrating Arduino with Python for real-time processing to improve navigation, control, and efficiency.
- Achieved a 20% reduction in route computation time, improving navigation accuracy, task efficiency, and overall system performance.

Technical Skills

Programming Languages: C, C++, Python
Machine Learning & AI Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, Stumpy, Pandas, NumPy, Matplotlib
Software Tools: MATLAB, GitHub, Jupyter Notebook, Google Colab, Visual Studio, MiniZinc (OR-Tools, Gurobi), Perforce
Certifications: Deep Learning: Image & Face Recognition, Data Structures (UCSD), Neural Networks & Deep Learning
Coursework: Algorithm Design & Analysis, Operating Systems, Machine Learning, Artificial Intelligence